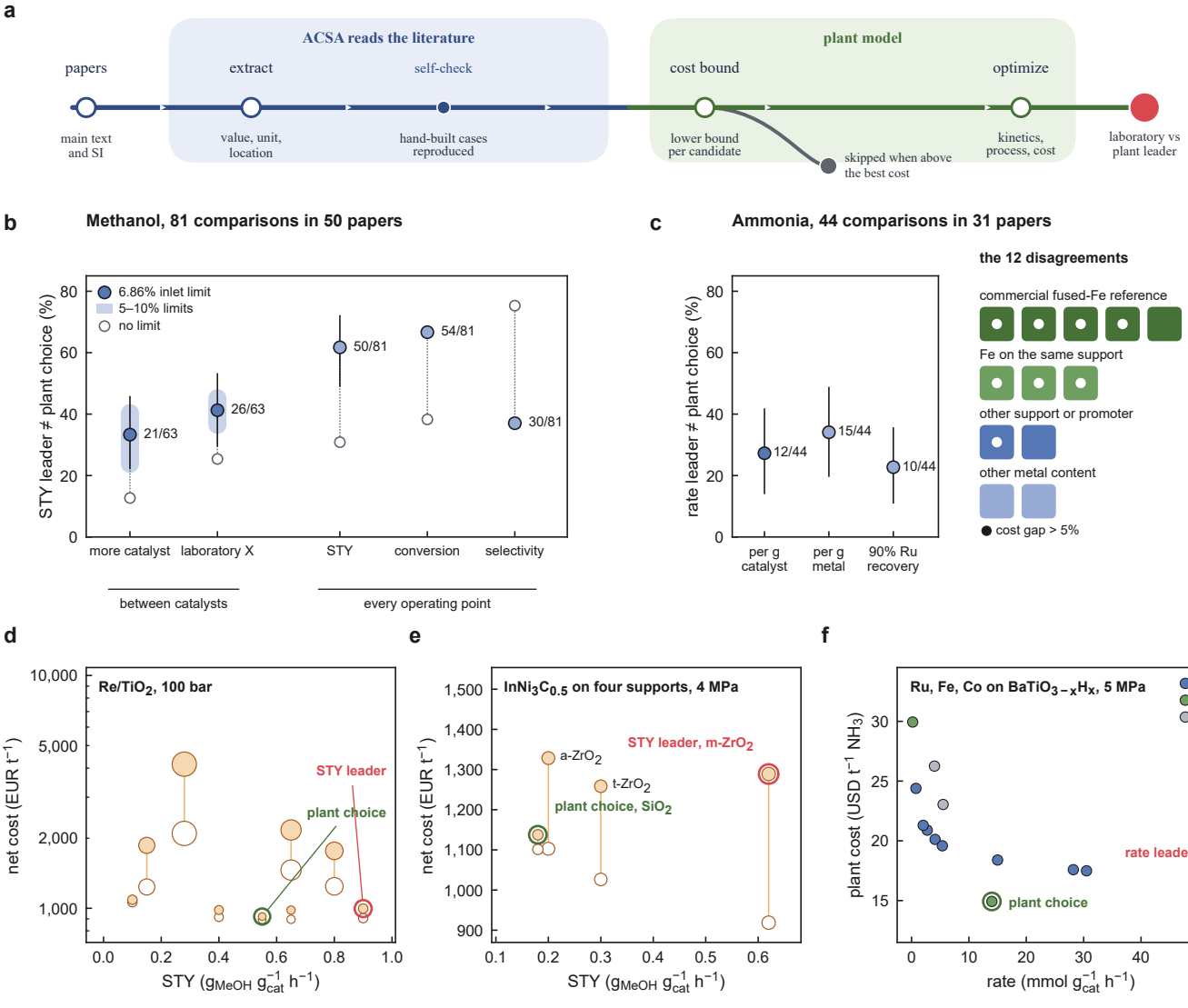

a
ACSA reads the literature
plant model
papers
extract
self-check
cost bound
optimize
main text and SI
value, unit, location
hand-built cases reproduced
lower bound per candidate
skipped when above the best cost
kinetics, process, cost
laboratory vs plant leader
b
Methanol, 81 comparisons in 50 papers
STY leader ≠ plant choice (%)
6.86% inlet limit
5–10% limits
no limit
21/63
26/63
50/81
54/81
30/81
more catalyst
laboratory X
STY
conversion
selectivity
between catalysts
every operating point
c
Ammonia, 44 comparisons in 31 papers
rate leader ≠ plant choice (%)
12/44
15/44
10/44
per g catalyst
per g metal
90% Ru recovery
the 12 disagreements
commercial fused-Fe reference
Fe on the same support
other support or promoter
other metal content
cost gap > 5%
d
Re/TiO2, 100 bar
net cost (EUR t−1)
STY leader
plant choice
STY (gMeOH gcat−1 h−1)
e
InNi3C0.5 on four supports, 4 MPa
net cost (EUR t−1)
a-ZrO2
t-ZrO2
STY leader, m-ZrO2
plant choice, SiO2
STY (gMeOH gcat−1 h−1)
f
Ru, Fe, Co on BaTiO3−xHx, 5 MPa
Ru
Fe
Co
plant cost (USD t−1 NH3)
rate leader
plant choice
rate (mmol gcat−1 h−1)